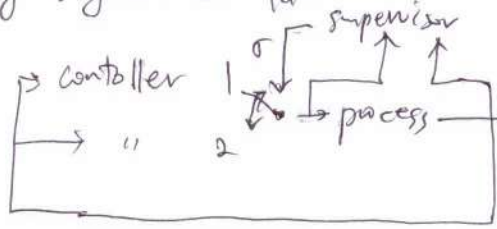


§ Ch1. Motivating Examples.

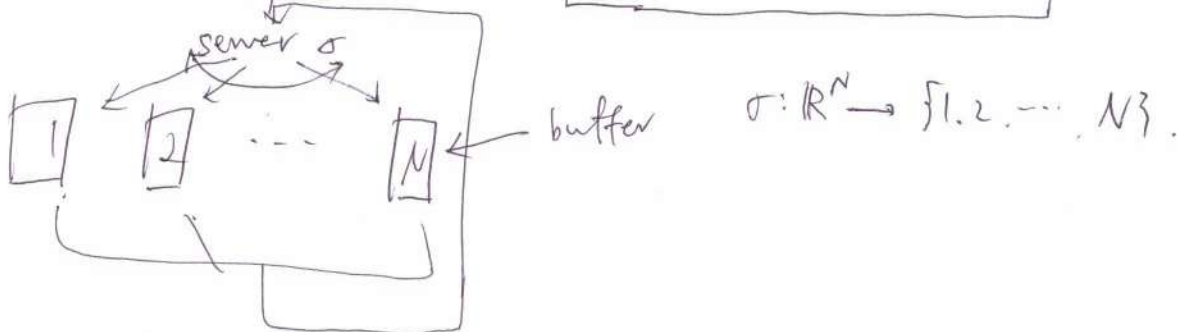
①

1.1 Switched Systems → switching signal or law

- Supervisory switching control



- Switched server system



- singular system with Markov switching

$\{r(t), t \geq 0\}$: Markov process
 $r: \mathbb{R}^+ \rightarrow \{1, 2, \dots, N\}$

1.2 Impulsive Systems

- SEIRS Model

$S(t)$: susceptible $E(t)$: exposed $I(t)$: infectious $R(t)$: recovered

$$N(t) = S(t) + E(t) + I(t) + R(t)$$

saturation incidence (hospital beds) or vaccination $\Rightarrow$ time delay or impulsive effects.

pg 4. equation: time delay and jumps (t^+)

- Insulin treatment (food $\Rightarrow$ sugar level continuously, insulin $\Rightarrow$ jump)

§ Ch2. Mathematical Background

2.1 Basic Definitions

$$\begin{cases} \dot{x} = f(t, x) \\ x(t_0) = x_0 \end{cases} \quad \text{IVP.} \quad \text{WLOG. } f(t, 0) = 0 \text{ so that } x=0: \text{equilibrium.}$$

Def 2.1

$x=0$ is (i) stable if $\forall \epsilon > 0 \exists \delta > 0$ s.t. $\|x_0\| < \delta \Rightarrow \|x(t)\| < \epsilon, \forall t \geq t_0$.

(ii) uniformly stable if stable + δ : indep of t_0 .

(iii) asymptotically stable if (i) + $\exists \bar{c} > 0$ s.t. $\|x_0\| < \bar{c} \Rightarrow \lim_{t \rightarrow \infty} x(t) = 0$.

(iv) uniformly asymptotically stable (ii) + c : indep of t_0 .

(v) exponentially stable $\|x(t)\| \leq k \|x_0\| e^{-\lambda(t-t_0)} \quad \forall \|x_0\| < c$.

(vi) unstable if (i) does not hold.

Def 2.2 $V: D \rightarrow \mathbb{R}$ is positive semi-definite if
 i) $V(t, 0) = 0$
 ii) $V(t, x) \geq 0 \quad \forall x \in D \setminus \{0\}$.

positive definite if i) $V(t, x) > 0$

radially unbounded: positive definite & $\lim_{\|x\| \rightarrow \infty} V(t, x) = \infty \quad \forall t$

Def 2.3 $D^+V(t, x) = \lim_{h \rightarrow 0^+} \sup \frac{1}{h} [V(t+h, x+hf(t, x)) - V(t, x)]$

Diri derivative

$$\dot{V}(t, x) = \frac{\partial V(t, x)}{\partial t} + \nabla_x V(t, x) \cdot f(t, x)$$

To guarantee stability, need special class of fns. ex) $k, k_L, k_1, k_2, \dots$

Thm 2.1 w_1, w_2 pos. def. V satisfies
 i) $w_1(x) \leq V(t, x) \leq w_2(x) \quad k_c, k_b, k_v, \dots$
 ii) $\dot{V}(t, x) = \frac{\partial V(t, x)}{\partial t} + \nabla_x V(t, x) \cdot f(t, x) \leq 0$
 $\Rightarrow x: \text{uniformly stable}$

If $\dot{V}(t, x) \leq -w_3(x) \quad \forall (t, x)$, then $x \equiv 0$ uniformly asymptotically stable

If $\exists r, c$ s.t. $\min_{\|x\|=r} w_1(x) > c$, ~~$w_2(x) \leq c$~~ for $\forall x$ starts in B_r s.t. $w_2(x) \leq c$, we have $\|x(t)\| \leq \beta(\|x(t_0)\|, t-t_0)$, $\beta \in k_L$

w_1 : radially unbounded $\Rightarrow x \equiv 0$ globally uniformly asymptotically stable

Note $k_L: \beta(\cdot, s) \in K$, $\beta(r, \cdot)$: decreasing & $\beta(r, s) \rightarrow 0$ as $s \rightarrow \infty$

K : ~~$\alpha \in K$~~ if α is strictly increasing & $\alpha(0) = 0$

-ISS (Input-to-State stable)

* $\begin{cases} \dot{x} = f(t, x, u) & t \geq t_0 \text{ perturbation} \\ x(t_0) = x_0 \end{cases}$
 input $u \in PC(\mathbb{R}_+; \mathbb{R}^n)$ (i.e. piecewise continuous)
 $\sup_{t \geq t_0} \|u(t)\| < \infty$

Def 2.11 (*) is ISS if $\exists \beta \in k_L$ & $r \in K$ s.t. $\forall x_0, \text{bdd } u$, $x(t)$ exists and $\|x(t)\| \leq \beta(\|x_0\|, t-t_0) + r\left(\sup_{t_0 \leq s \leq t} \|u(s)\|\right) \quad \forall t \geq t_0$

Remark $u \equiv 0 \Rightarrow$ ISS becomes globally uniformly asymptotic stable

Thm 2.2 $a, b \in K_\infty$ ($a, b \in K$ & $a(r), b(r) \rightarrow \infty$ as $r \rightarrow \infty$)

$c \in K$, c : pos def

V satisfies $\begin{cases} b(\|x\|) \leq V(t, x) \leq a(\|x\|) & \forall (t, x) \\ \dot{V}(t, x, u) \leq -c(x) & \forall \|x\| \geq \rho(\|u\|) \end{cases}$

for $\forall (t, x, u) \in \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^m$

$\Rightarrow$ system is ISS with $\gamma(\cdot) = b^{-1}(a(\rho(\cdot)))$

2.2 Comparison Method

Thm 2.3 $\begin{cases} \dot{x} = f(t, x) \\ x(t_0) = x_0 \end{cases}$ $V: \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}_+$ conti, ~~loc~~ loc Lipschitz in x .

Assume V satisfies $D^+V(t, x) \leq g(t, V(t, x))$ where g : conti.

Let $r(t) = r(t; t_0, u_0)$ be maximal solution of $\begin{cases} \dot{u} = g(t, u) \\ u(t_0) = u_0 \geq 0 \end{cases}$

Then $V(t_0, x_0) \leq u_0$ implies $V(t, x(t)) \leq r(t) \quad \forall t \geq t_0$. $f(t, x) \leq g(t, x)$
 $\Rightarrow V(x) \leq r(t)$
 $\Rightarrow x(t) \leq r(t)$

Thm 2.4 $a, b \in \mathbb{K}$, V : loc lip in x , (i) $b(\|x\|) \leq V(t, x) \leq a(\|x\|) \quad \forall(t, x)$
(ii) $D^+V(t, x) \leq g(t, V(t, x))$
where g : conti & $g(t, 0) = 0 \quad \forall t$.

Then stability property of $u=0$ implies stab prop of $x=0$.

2.3 Delay Systems

$$C_r = C([-r, 0]: \mathbb{R}^n)$$

Def 2.12 Let $t^* \in \mathbb{R}$, $a > 0$. If x is a fn mapping $[t^*-r, t^*+a] \rightarrow \mathbb{R}^n$, for each $t \in [t^*, t^*+a]$, define $x_t: [-r, 0] \rightarrow \mathbb{R}^n$ by $x_t(s) = x(t+s)$
 $\|x_t\|_r = \sup_{t-r \leq \theta \leq t} \|x(\theta)\|$

$\begin{cases} \dot{x}(t) = f(t, x_t) \\ x_{t_0} = \phi(s) \end{cases}$: functional differential equation, f : functional operator.
↪ IVP of delay-type system.
 $s = -r$, $x_t(s) = x(t-r)$: delay systems.

Def 2.13, 2.14 skip.

Thm 2.5 stability of time-delay (add some conditions such as $V(t+\theta, \psi(\theta)) \leq V(t, \psi(0)) \dots$)

* Important Example

$$\dot{x}(t) = Ax(t) + Bx(t-r) \quad A, B: \text{matrices} \quad r: \text{time-delay}$$

⇒ One encounters $\dot{V}(t) \leq -\alpha V(t) + \beta \sup_{t-r \leq s \leq t} V(s)$, $\alpha, \beta > 0$

Upper bound of $V(t)$.

Lemma 2.1 $\alpha, \beta > 0 \Rightarrow \exists \gamma > 0, k > 0$ s.t. $V(t) \leq ke^{-\gamma(t-t_0)}$ where $-\gamma = -\alpha + \beta e^{\gamma r}$
 $k = \max_{t_0-r \leq s \leq t_0} V(s)$
LH 2.1 Lemma: skip.

2.4 Impulsive Systems

(4)

$$\dot{x}(t) = f(t, x(t)) \quad t \neq t_k(x) \quad \left\{ \begin{array}{l} x(t) = x(t+) - x(t) \\ x(t+) = \lim_{\varepsilon \rightarrow 0+} x(t+\varepsilon) \end{array} \right. , \quad x(t+) = \lim_{\varepsilon \rightarrow 0+} x(t+\varepsilon)$$

$$\Delta x(t) = F(t, x(t)) \quad t = t_k(x) \quad x(t) = x(t-) \quad (LC)$$

$$x(t_0+) = x_0$$

$$0 = z_0(x) < z_1(x) < z_2(x) < \dots, \quad \lim_{k \rightarrow \infty} z_k(x) = \infty$$

$$\text{At } t = z_k(x), \quad x(t+) = x(t) + F(t, x(t))$$

To have RC sol,

$$\begin{cases} \dot{x}(t) = f(t, x(t)) & t \neq z_k(x(t-)) \\ \Delta x(t) = F(t, x(t-)) & t = z_k(x(t-)) \\ x(t_0) = x_0 \end{cases}$$

$z_k(x)$: impulse time. ~~(finite or countable)~~
if z_k : constants,
then all solution undergo impulses
at same time.

$$x = x(t; t_0, x_0) = \begin{cases} x(t; t_0, x_0) & t_0 < t \leq t_1 \\ x(t; t_1, x(t_1+)) & t_1 < t \leq t_2 \\ \vdots \\ x(t; t_k, x(t_k+)) & t_k < t \leq t_{k+1} \end{cases}$$

Stability: only for $z_k(x) = z_k \quad \forall x$

Def 2.16 skip

$$(2.24) \quad \begin{cases} \dot{x}(t) = Ax(t) & t \neq t_k \quad k \in \mathbb{N} \\ \Delta x(t) = B_k x(t) & t = t_k \\ x(t_0+) = x_0 \end{cases}$$

$$(A1) \exists 0 < \rho_1 \leq \rho \text{ s.t. } \forall z_k \in \mathbb{R}_+, \quad \|x(z_k-)\| < \rho_1 \Rightarrow \|x(z_k)\| < \rho$$

x defined on $P(\mathbb{R}_+, D)$

$$(A2) \forall k \in \mathbb{N}, \quad z_{\sup} = \sup \{z_k - z_{k-1}\} < \infty$$

$$z_{\inf} = \inf \{z_k - z_{k-1}\} > 0$$

Thm 2.6 A : Hurwitz. (2.24) is globally exponentially stable if

$$\forall k \in \mathbb{N} \quad \ln d_k - \gamma(t_k - t_{k-1}) \leq 0 \quad \forall k \in \mathbb{N}, \quad d_k = \frac{\lambda_{\max}((I+B_k)^T P (I+B_k))}{\lambda_{\min}(P)}$$

$$0 < \gamma < \xi, \quad \xi = \frac{\lambda_{\min}(Q)}{\lambda_{\min}(P)}$$

$$Q: \text{pos. def.} \quad P: \text{pos def.}, \quad A^T P + P A = -Q$$

$$(pf) \quad V(x) = x^T P x, \quad v(t) = V(x(t)) \quad \frac{\partial}{\partial x} V(x) = x^T (P^T + P)$$

$$\Rightarrow \dot{v}(t) \leq -\xi v(t), \quad t \in (t_{k-1}, t_k], \quad \xi = \lambda_{\min}(Q) / \lambda_{\max}(P)$$

$$v(t) \leq v(t_{k-1}^+) e^{-\xi(t-t_{k-1})} \quad t \in (t_{k-1}, t_k]$$

$$v(t_k^+) = x(t_k^+)^T P x(t_k^+) = x(t_k^+)^T (I+B_k)^T P (I+B_k) x(t_k)$$

$$\leq \lambda_{\max}((I+B_k)^T P (I+B_k)) x(t_k)^T x(t_k) = d_k v(t_k)$$

* Hurwitz

$$\begin{pmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ & a_1 & a_3 \\ & & a_2 \\ & & & a_1 \\ & & & & a_n \\ & & & & & a_{n-1} \\ & & & & & & a_{n-2} \end{pmatrix}$$

$$v(t_k) = x(t_k)^T P x(t_k) \geq \lambda_{\min} x(t_k)^T x(t_k)$$

$$V(t) \leq V(t_0^+) e^{-\beta(t-t_0)} \quad t \in (t_0, t_1]$$

$$v(t_1^+) \leq \alpha_1 v(t_1) \leq \alpha_1 V(t_0^+) e^{-\beta(t_1-t_0)}$$

$$V(t) \leq V(t_0^+) \alpha_1 e^{-\beta(t_1-t_0)} e^{-\beta(t-t_1)} = V(t_0^+) \alpha_1 e^{-\beta(t-t_0)} \quad t \in (t_1, t_2]$$

Generally, for $t \in (t_k, t_{k+1}]$, $V(t) \leq V(t_0^+) \alpha_1 \alpha_2 \dots \alpha_k e^{-\beta(t-t_0)} e^{-V(t-t_0)} e^{-(\beta-\nu)(t-t_0)}$

$$= V(t_0^+) \alpha_1 \alpha_2 \dots \alpha_k \cancel{e^{-V(t-t_0)}} \cancel{e^{-(\beta-\nu)(t-t_0)}} e^{-\nu(t-t_0)}$$

$$= V(t_0^+) \alpha_1 e^{-\nu(t_1-t_0)} \dots \alpha_k e^{-\nu(t_k-t_{k-1})} e^{-(\beta-\nu)(t-t_k)}$$

$$\leq V(t_0^+) e^{-(\beta-\nu)(t-t_0)} \quad t \geq t_0$$

$$\Rightarrow \|x(t)\| \leq K \|x(t_0^+)\| e^{-(\beta-\nu)(t-t_0)/2}$$

$$(\lambda_{\min}(P) \|x(t)\|)^2 \leq x(t)^T P x(t) \leq x(t_0^+)^T P x(t_0^+) e^{-(\beta-\nu)(t-t_0)} \leq \lambda_{\max}(P) \|x(t_0^+)\|^2 e^{-(\beta-\nu)(t-t_0)}$$

$$K = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}}$$

2.5 Comparison Method for Impulsive Systems

Thm 2.7 (Comparison Thm)

V : conti in $(t_{k-1}, t_k] \times \mathbb{R}^n$, $\lim_{(t,x) \rightarrow (t_k^+, x)} V(t,x) = V(t_k^+, x)$ loc Lipschitz in x .

$$D^+ V(t,x) \leq g(t, V(t,x)) \quad t \neq t_k,$$

$$V(t, x + F(t, x)) \leq \psi_k(V(t, x)) \quad t = t_k$$

g : conti in $(t_{k-1}, t_k] \times \mathbb{R}_+$, $\lim_{(t,y) \rightarrow (t_k^+, x)} g(t,y) = g(t_k^+, x)$, ψ_k : nondecreasing.

$v(t)$: max sol of

$$\begin{cases} \dot{u}(t) = g(t, u) & t \neq t_k \\ \Delta u(t) = \psi_k(u) & t = t_k \\ u(t_0^+) = u_0 \geq 0 & t_0 \geq 0 \end{cases}$$

Then,

$$V(t_0^+, x_0) \leq u_0 \text{ implies } V(t, x(t)) \leq v(t) \quad \forall t \geq t_0$$

Thm 2.8 (Stability Thm) ...

2.6 Impulse systems with time delay

$p \in [a, b]$

read ...

2.1 Stochastic Differential Equations

$$X_n \rightarrow X \text{ a.s.} : P(\{\omega \mid \lim_{k \rightarrow \infty} X_k(\omega) = X(\omega)\}) = 1$$

$$\text{in prob} : \forall \varepsilon > 0, \lim_{k \rightarrow \infty} P(\{\omega \mid |X_k(\omega) - X(\omega)| > \varepsilon\}) = 0$$

$$\text{in } L^p : \lim_{k \rightarrow \infty} E|X_k - X|^p = 0$$

$$\text{in distribution} : \lim_{k \rightarrow \infty} F_k(x) = F(x) \quad F_k, F : \text{distribution ftn.}$$

$$P(\{\omega \mid X(\omega) \leq x\})$$

a.s. $\Rightarrow$ in prob $\Rightarrow$ in dist.
 $L^p \Rightarrow$

stochastic process: $\{X_t(\omega) : t \in I, \omega \in \Omega\}$

filtration $(\mathcal{F}_t)_{t \geq 0} : \mathcal{F}_t \subset \mathcal{F}_s \text{ for } t \leq s$

$\{\mathcal{F}_t\}_{t \geq 0}$ is right conti if $\mathcal{F}_t = \bigcap_{s > t} \mathcal{F}_s$

usual conditions = right conti, $\mathcal{F}_0$ contains all P -null sets.

$\{X_t\}$: conti = continuous for almost all $\omega \in \Omega$.

cadlag = right-conti, $\exists \lim_{s \uparrow t} X_s(\omega)$ a.s.

$\mathcal{F}_t$ -adapted ~~to $\mathcal{F}_t$~~ : X_t is $\mathcal{F}_t$ m'ble.
 X_t, Y_t are indistinguishable if $P(\{\omega \mid X_t(\omega) = Y_t(\omega), \forall t \in \mathbb{R}_+\}) = 1$

D : open subset. first exit time $\tau = \inf\{t \in \mathbb{R}_+ \mid X(t) \notin D\}$

Def 2.23 moment

$$m_k(t) = E[X^k(t)], \quad \text{Var}(X(t)) = m_2(t) - m_1^2(t)$$

Def 2.24 $(\Omega, \mathcal{F}, P)$ prob space

(W_t) : Wiener process if (i) conti

(ii) $P(W_0 = 0) = 1$

(iii) $\forall 0 \leq s < t, W_t - W_s \perp \mathcal{F}_s$

(iv) ~~$W_t - W_s \sim N(0, t-s)$~~ $E(W_{t+h} - W_t) = \mu h$
 $E((W_{t+h} - W_t)^2) = \sigma^2 h$

$\mu = 0, \sigma^2 = 1 \Rightarrow$ standard Wiener process

autocorrelation of $X : R(t_1, t_2) = E(X_{t_1} X_{t_2}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f(x_1, t_1; x_2, t_2) dx_1 dx_2$

$X(t)$: stationary iff $\forall t_1, \dots, t_n, \tau, f_{X(t_1), \dots, X(t_n)}(x_1, \dots, x_n) = f_{X(t_1+\tau), \dots, X(t_n+\tau)}(x_1, \dots, x_n)$

Def 2.25 A stochastic process N is Gaussian white noise process iff

it is a stationary Gaussian process with mean zero and

$$R(\tau) = C\delta(\tau), \quad \delta : \text{Dirac delta}$$

$$= E(X_{t+\tau} X_t)$$

$$\text{rank Var}(N(t)) = \infty$$